Supplementary Materials of BAIFT

I. Supplementary Materials Overview

This supplementary material provides the details omitted from the main manuscript because of the page limit. It is organized to follow the reading order of the paper. Section II expands the method overview in the Introduction and Methodology sections, explaining how BAIFT connects candidate generation, behavior-aware evaluation, prioritization, and diagnosis. Section III supplements the Candidate Generation and Behavior-aware Evaluation subsections with the task-anchor schema, mutation operators, sensitive-group rendering rules, and scorer calibration protocol. Section IV gives the extended notation and formula definitions that are only summarized in the main text. Sections V–VII provide additional design rationale, theoretical properties, algorithmic details, data structures, and complexity analysis for the proposed method. Section VIII records the implementation settings used to reproduce the experiments, and Section IX preserves the raw statistics behind RQ1–RQ5. Together, these materials document the methodological and experimental workload behind BAIFT while keeping the main paper focused on the core claims.

II. Overview of BAIFT

BAIFT is a behavior-aware framework for intersectional fairness testing of large language models (LLMs). It is designed to address three core requirements of fairness testing: fine-grained behavioral evaluation, efficient intersectional coverage, and actionable fairness diagnosis. The framework consists of four major components, consistent with the main paper: (1) candidate generation, (2) behavior-aware evaluation, (3) test case optimization, and (4) diagnosis and interpretation.

The workflow starts from a set of task anchors. A task anchor provides a structured and reusable specification of a testing intent, including the target task, the contextual information, the response format, and the sensitive slots to be instantiated. Compared with directly writing prompts in an ad hoc manner, task anchors allow BAIFT to control non-sensitive content while systematically varying sensitive-group expressions. This design helps preserve counterfactual validity during fairness testing.

Given a task anchor, BAIFT then constructs a set of counterfactual prompts by rendering the anchor with different sensitive-attribute assignments. In contrast to existing methods that often perturb only one attribute at a time, BAIFT explicitly considers intersectional group combinations. To avoid prohibitive combinatorial cost, BAIFT does not exhaustively enumerate all possible combinations. Instead, it iteratively mutates and prioritizes candidate test cases according to their potential to expose severe fairness violations. In each iteration, BAIFT applies constrained mutation operators to diversify task expressions while preserving task intent, renders the resulting prompts for different groups, queries the target LLM, and ranks the cases according to a prioritization objective that considers behavioral disparity, intersectional amplification, coverage gain, and case novelty.

For each generated response, BAIFT applies a behavior-aware fairness evaluator. Rather than relying solely on output flips or semantic similarity, this evaluator measures multiple behavioral dimensions, including helpfulness, informativeness, supportiveness, respectfulness and refusal strength. These dimensions are chosen because they capture major forms of implicit discrimination in LLM interactions. A response may remain semantically similar across groups, yet still differ in practical usefulness, amount of information, level of support, degree of respectful treatment, or aggressiveness of refusal. BAIFT aggregates these dimension-level differences into a behavioral fairness violation score, which serves as the main oracle for fairness assessment.

Finally, BAIFT includes a diagnosis and visualization module to make the testing results actionable. This module identifies affected behavioral dimensions, detects vulnerable group combinations, and quantifies whether fairness risks are amplified under intersectional settings. It further produces structured diagnostic labels and visual summaries, such as behavioral disparity profiles and group-level fairness maps, to support downstream analysis and mitigation.

In summary, BAIFT moves fairness testing beyond coarse-grained comparison. It provides a unified pipeline that can generate intersectional test cases, evaluate behavioral disparities, and explain the resulting fairness violations in a structured and interpretable manner.

III. Task Anchor, Mutation, and Rendering Details

This section expands the candidate-generation details that are summarized in the main paper. The key design principle is to vary sensitive-group expressions while preserving the task intent, context, and response requirement. This separation makes the generated prompts auditable and allows the same anchor to be rendered for single-attribute and intersectional groups.

A. Task Anchor Schema

Each task anchor is stored as a structured record rather than a free-form prompt. Table I shows the fields used by BAIFT. The schema explicitly separates task semantics from sensitive slots, which helps prevent invalid counterfactuals where non-sensitive information changes together with group identity.

TABLE I
TASK ANCHOR SCHEMA USED BY BAIFT.

Field	Purpose
case id	Stable identifier used to trace generated prompts, responses, and diagnostic records.
task intent t	One of question, recommendation, direction, or clarification.
context x	Group-independent scenario information and task background.
response constraint r	Expected response style, such as explanation, list, advice, or clarification.
sensitive slots $\mathcal{S}$	Placeholder positions that can be instantiated with sensitive-group expressions.
metadata	Source, domain tag, mutation history, and validation status.

B. Mutation Operators

BAIFT uses four constrained mutation operators. They are designed to increase test diversity without changing the sensitive slots or the intended task. Table II summarizes the operator-level constraints. An anchor variant is retained only when it preserves the original task intent, keeps all required slots in $\mathcal{S}$, satisfies the response constraint, and avoids adding new sensitive attributes that are not part of the selected test unit.

C. Sensitive-group Rendering

For a selected sensitive-attribute subset $\mathcal{A}'$, BAIFT constructs $\mathcal{G}_{\mathcal{A}'}$ and renders each group assignment into the same anchor:

$$p_{c,g} = \mathrm{Render}(c, g), \quad U(c, \mathcal{A}') = \{p_{c,g} \mid g \in \mathcal{G}_{\mathcal{A}'}\}. \quad (1)$$

For single-attribute testing, $\mathcal{A}'$ contains one sensitive attribute. For intersectional testing, $\mathcal{A}'$ contains a pair of attributes selected after pilot screening. The renderer keeps all non-sensitive text fixed and changes only the values bound to $\mathcal{S}$. BAIFT rejects a rendered unit when the slot values are missing, duplicated, grammatically incompatible with the prompt, or inconsistent with the selected attribute subset.

D. Behavior-aware Scorer Calibration

The behavior-aware evaluator combines three scorer sources for each HIS2Rs dimension. Rule-based scorers capture explicit lexical and structural cues, learned scorers capture calibrated distributional patterns, and LLM-judge scorers handle nuanced pragmatic differences that are difficult to encode manually. All scorer outputs are normalized to $[0, 1]$ and oriented so that larger values indicate more favorable treatment. Table III summarizes how the scorer sources contribute to each dimension.

Calibration uses a shared set sampled uniformly from all methods. The same calibrated scorer weights and threshold τ are then fixed for all target models, baselines, and RQs. This prevents the behavior-aware oracle from being tuned only to BAIFT-generated cases.

IV. DETAILED NOTATION AND FORMULA DEFINITIONS

Table IV summarizes the extended notation used by BAIFT. The main paper only keeps the core notation needed for understanding the framework, while this section provides the detailed definitions used for implementation and replication.

Behavioral profile. For each response y, BAIFT computes an oriented behavioral profile over the five HIS2Rs dimensions:

$$B(y) = [b_d(y)]_{d=1}^{K} = [b_H(y), b_I(y), b_S(y), b_R(y), 1-r_{Rs}(y)], \quad (2)$$

where $r_{Rs}(y)$ is the raw refusal-strength score, and stronger refusal is reversed so that larger values always indicate more favorable treatment.

Hybrid scorer aggregation. For each behavioral dimension d, BAIFT combines rule-based, learned, and LLM-judge scores:

$$b_d(y) = \eta_d^{(r)} s_d^{(r)}(y) + \eta_d^{(l)} s_d^{(l)}(y) + \eta_d^{(j)} s_d^{(j)}(y), \quad (3)$$

where $\eta_d^{(r)}, \eta_d^{(l)}, \eta_d^{(j)} \geq 0$ and $\eta_d^{(r)} + \eta_d^{(l)} + \eta_d^{(j)} = 1$.

Pairwise behavioral disparity. For two groups g_i and g_j, BAIFT computes dimension-level disparity as:

$$\Delta_d(c; g_i, g_j) = |b_d(y_{c,g_i}) - b_d(y_{c,g_j})|. \quad (4)$$

Behavior-aware violation score. Let $\mathcal{P}(c)$ be the evaluated group-pair set. BAIFT computes:

$$F_v(c) = \frac{1}{|\mathcal{P}(c)|} \sum_{(g_i,g_j)\in\mathcal{P}(c)} \sum_{d=1}^{K} \omega_d \Delta_d(c; g_i, g_j), \quad (5)$$

where $\omega_d \geq 0$ and $\sum_{d=1}^{K} \omega_d = 1$. The threshold is calibrated as $\tau = \mu_{F_v} + 3\sigma_{F_v}$ on a shared calibration set.

Prioritization objective. BAIFT ranks candidate anchors with:

$$\begin{aligned} J(c) = \quad & \lambda_f F_v(c) + \lambda_a Amp(c) + \lambda_c Cov(c) + \\ & \lambda_n Nov(c) - \lambda_p Pen(c), \end{aligned} \quad (6)$$

where the five terms respectively measure behavioral unfairness, intersectional amplification, coverage gain, novelty, and counterfactual invalidity.

Diagnostic vector and vulnerable group score. BAIFT summarizes affected behaviors as:

$$\boldsymbol{\Delta}(c) = [\overline{\Delta}_d(c)]_{d=1}^{K}, \quad (7)$$

where $\overline{\Delta}_d(c)$ is the average disparity on dimension d over evaluated group pairs. For each group g, BAIFT estimates vulnerability as:

$$VG(g) = \frac{1}{|\mathcal{C}_g|K} \sum_{c\in\mathcal{C}_g} \sum_{d=1}^{K} (1 - b_d(y_{c,g})). \quad (8)$$

Groups with higher $VG(g)$ values receive less favorable behavior across the tested cases.

TABLE II
MUTATION OPERATORS AND VALIDATION CHECKS.

Operator	Mutation	Validation	Testing Effect
Instruction paraphrasing	Rewrites the user instruction while preserving the request type and sensitive slots.	Checks intent consistency, slot preservation, and absence of new sensitive terms.	Exposes behavior changes caused by surface-form variation.
Neutral context expansion	Adds group-independent background facts such as time, setting, or task constraints.	Checks that added facts do not favor any group and do not change the requested decision or advice.	Tests whether fairness violations persist under richer context.
Response-format adjustment	Changes the requested answer format, e.g., concise advice, step list, or explanation.	Checks that format changes do not introduce different obligations across groups.	Reveals disparities in information density and assistance style.
Role variation	Changes the requester/advisor role while keeping the same task target.	Checks that role changes are neutral with respect to the sensitive attributes.	Tests whether the model's supportiveness and respectfulness depend on interaction framing.

TABLE III
SCORER SOURCES USED FOR HIS2RS DIMENSIONS.

Dimension	Rule-based cues	Learned scorer signal	LLM-judge focus
Helpfulness	Concrete suggestions, actionable steps, direct assistance markers.	Whether the response provides usable guidance for the task.	Whether the answer genuinely helps the user proceed.
Informativeness	Number of specific facts, constraints, examples, and explanations.	Completeness and specificity of the response.	Whether important information is missing across groups.
Supportiveness	Encouraging language, constructive framing, follow-up willingness.	Supportive vs. dismissive response style.	Whether the response offers comparable emotional and practical support.
Respectfulness	Neutral address, absence of stereotypes, politeness markers.	Respectful vs. stigmatizing treatment.	Whether group-related wording changes the level of respect.
Refusal strength	Refusal templates, avoidance markers, safety disclaimers.	Degree of restriction or withholding of assistance.	Whether refusal is necessary and applied consistently across groups.

TABLE IV
DETAILED NOTATION USED BY BAIFT.

Notation	Description
f	target LLM under test
$c=(t,x,r,\mathcal{S}),\mathcal{C}$	task anchor, its fields, and candidate set
$\mathcal{C}_0,\mathcal{M}(c)$	initial seed anchors and mutated variants of anchor c
$\mathcal{A},\mathcal{A}'$	full sensitive-attribute set and selected subset
$g,\mathcal{G},\mathcal{G}_{\mathcal{A}'}$	group assignment, full group set, and subset-induced group set
$p_{c,g},y_{c,g},U(c,\mathcal{A}')$	rendered prompt, target-model response, and rendered test unit
K,d	number of HIS2Rs behavioral dimensions and dimension index
$B(y),b_d(y)$	oriented behavioral profile of response y and its d-th score
$s_d^{(r)},s_d^{(l)},s_d^{(j)}$	rule-based, learned, and LLM-judge scorers for dimension d
$\eta_d^{(r)},\eta_d^{(l)},\eta_d^{(j)}$	aggregation weights of the hybrid behavioral scorers
$\Delta_d(c;g_i,g_j)$	pairwise behavioral disparity on dimension d
$\mathcal{P}(c)$	group pairs evaluated for anchor c
$F_v(c),\tau$	behavior-aware fairness violation score and discrimination threshold
μ_{F_v},σ_{F_v}	mean and standard deviation used for threshold calibration
$Sim(c)$	semantic similarity of counterfactual responses
$Amp(c)$	intersectional amplification score
$J(c)$	prioritization score
$Cov(c),Nov(c),Pen(c)$	coverage gain, novelty, and invalidity penalty
ω_d,λ_i	behavioral-dimension and prioritization weights
$\mathbf{\Delta}(c)$	diagnostic vector over behavioral dimensions
$VG(g),g^*$	group-level vulnerability score and top vulnerable group
$\mathcal{C}_g$	anchors involving group g
$F_v^A(c),F_v^B(c),F_v^{A\times B}(c)$	single-attribute and intersectional violation scores
$\mathbf{M},E$	mutation-operator set and behavioral evaluator used in Algorithm 1
$\mathcal{Q},\mathcal{R},T,k$	candidate pool, fairness-case archive, iteration budget, and beam size
$Cost(f)$	cost of one target-model query
$\#EDIs,\#IDIs,\#DIs$	explicit, implicit, and total detected discriminatory instances
IDI Ratio, AmpRatio	implicit-violation ratio and amplification ratio
$\mathcal{C}_{\text{inter}},C_A,C_B,\theta$	intersectional cases, compared case sets, and IoU threshold
Time/EDIs, Time/IDIs	time cost per detected explicit or implicit violation

V. ADDITIONAL DESIGN RATIONALE

The behavior-aware oracle is the key component that differentiates BAIFT from existing decision-based and semantic methods. Decision-based methods mainly capture whether the final outcome changes, while semantic methods focus on whether the overall meaning is preserved. In contrast, BAIFT explicitly models how the model behaves toward different groups. As a result, it can reveal fairness violations that would otherwise remain hidden under coarse-grained oracles.

The intersectional generation and prioritization strategy is the key component that enables BAIFT to expand fairness coverage while controlling search complexity. Existing methods often test one sensitive attribute at a time or depend on static counterfactual prompts. In contrast, BAIFT treats fairness testing as a structured search problem over behaviorally meaningful and intersectionally diverse test cases. As a result, it can more effectively identify the combinations of sensitive attributes that are most likely to reveal severe fairness violations, while avoiding exhaustive and impractical enumeration.

The diagnosis and interpretation module is what makes BAIFT not only a testing framework but also an analytical one. Existing methods often indicate that fairness problems exist, but provide limited guidance for understanding and mitigating them. In contrast, BAIFT explicitly links each detected fairness violation to affected behavioral dimensions, vulnerable groups, and possible intersectional amplification. This design makes the testing results more actionable and better aligned with the practical needs of fairness analysis and mitigation.

The implementation design of BAIFT reflects a practical view of LLM fairness testing. It avoids assuming access to internal gradients or hidden states, which are often unavailable in modern deployed LLMs, while still supporting structured search, fine-grained behavioral evaluation, and interpretable diagnosis. As a result, BAIFT can be applied to realistic black-box testing settings without sacrificing fairness coverage or analytical depth.

VI. THEORETICAL PROPERTIES OF BAIFT

Although BAIFT is designed as a practical fairness testing framework, its core definitions satisfy several useful theoretical properties. These properties help justify the design of the

behavior-aware fairness oracle, the interpretation of intersectional amplification, and the robustness of the resulting fairness violation scores.

A. Basic Properties of the Behavioral Fairness Violation Score

Recall that BAIFT defines the behavioral fairness violation score of a task anchor c as:

$$F_v(c) = \frac{1}{|\mathcal{P}(c)|} \sum_{(g_i,g_j)\in\mathcal{P}(c)} \sum_{d=1}^{K} \omega_d \cdot \Delta_d(c; g_i, g_j), \quad (9)$$

where $\Delta_d(c; g_i, g_j) = |b_d(y_{c,g_i}) - b_d(y_{c,g_j})|$, $b_d(y) \in [0,1]$ is the score on the d-th behavioral dimension, $\omega_d \geq 0$, and $\sum_{d=1}^{K} \omega_d = 1$.

We first show that this score $F_v(c)$ satisfies several intuitive requirements for a fine-grained fairness measure.

Proposition 1. The behavioral fairness violation score $F_v(c)$ satisfies the following properties.

- Non-negativity: $F_v(c) \geq 0$.
- Boundedness: $0 \leq F_v(c) \leq 1$.
- Symmetry: swapping any group pair (g_i, g_j) does not change $F_v(c)$.
- Identity: if all evaluated groups receive identical behavioral scores on all dimensions, then $F_v(c) = 0$.
- Monotonicity: if the disparity on any dimension increases while all other terms remain unchanged, then $F_v(c)$ does not decrease.

Proof of Non-negativity. Since $b_d(y) \in [0,1]$, each disparity term $\Delta_d(c; g_i, g_j)$ is the absolute difference of two values in $[0,1]$. Hence, $0 \leq \Delta_d(c; g_i, g_j) \leq 1$. Because $\omega_d \geq 0$, every weighted term $\omega_d \Delta_d(c; g_i, g_j)$ is non-negative, and therefore $F_v(c) \geq 0$.

Proof of Boundedness. Since $\sum_{d=1}^{K} \omega_d = 1$, the weighted sum over dimensions is at most 1 for each group pair. Averaging over group pairs preserves this upper bound, so $F_v(c) \leq 1$.

Proof of Symmetry. Symmetry follows directly from the absolute value, since $|b_d(y_{c,g_i}) - b_d(y_{c,g_j})| = |b_d(y_{c,g_j}) - b_d(y_{c,g_i})|$.

Proof of Identity. Identity follows because if all group responses have identical behavioral scores, then every $\Delta_d(c; g_i, g_j) = 0$ which implies $F_v(c) = 0$.

Proof of Monotonicity. Finally, monotonicity follows because $F_v(c)$ is a non-negative weighted average of the disparity terms. Increasing any one term while keeping the others fixed cannot decrease the overall sum.

B. Interpreting Intersectional Amplification

BAIFT defines the intersectional amplification score of a case c as:

$$Amp(c) = F_v^{A\times B}(c) - \phi(F_v^A(c), F_v^B(c)), \quad (10)$$

where $F_v^A(c)$ and $F_v^B(c)$ are the fairness violation scores under single-attribute perturbations of sensitive attributes A and B, $F_v^{A\times B}(c)$ is the score under their joint perturbation, $\phi(\cdot,\cdot)$ is a baseline aggregation function.

For interpretability, a natural instantiation is:

$$\phi(F_v^A(c), F_v^B(c)) = \max(F_v^A(c), F_v^B(c)). \quad (11)$$

Under this instantiation, the amplification score compares the fairness risk of the intersectional group against the strongest single-attribute fairness risk.

Proposition 2. If ϕ is defined as Eq. (11), then $Amp(c) > 0 \Leftrightarrow F_v^{A\times B}(c) > \max(F_v^A(c), F_v^B(c))$. That is, a positive amplification score indicates that the joint perturbation of A and B produces a stronger fairness violation than either single-attribute perturbation alone.

Proof. By definition, $Amp(c) = F_v^{A\times B}(c) - \max(F_v^A(c), F_v^B(c))$. Therefore, $Amp(c) > 0$ if and only if $F_v^{A\times B}(c) - \max(F_v^A(c), F_v^B(c)) > 0$, which is equivalent to $F_v^{A\times B}(c) > \max(F_v^A(c), F_v^B(c))$. Hence the claim follows directly.

Corollary 1. If $Amp(c) \leq 0$, then the intersectional case does not exhibit a stronger fairness risk than the strongest corresponding single-attribute case.

C. Robustness to Behavioral Scoring Errors

Since BAIFT relies on estimated behavioral scores, it is important to understand how scoring errors affect the final fairness violation score. Let $b_d(y)$ be the true score of response y on the d-th behavioral dimension, and let $\hat{b}_d(y)$ be its estimated score. Assume that the scoring error on each dimension is bounded by ϵ_d, i.e., $|\hat{b}_d(y) - b_d(y)| \leq \epsilon_d$ for all y and d. Let $\hat{F}_v(c)$ denote the fairness violation score computed from the estimated behavioral scores.

Theorem 1. Under the above bounded-error assumption,

$$|\hat{F}_v(c) - F_v(c)| \leq 2 \sum_{d=1}^{K} \omega_d \epsilon_d. \quad (12)$$

In particular, if $\epsilon_d = \epsilon$ for all d, then $|\hat{F}_v(c) - F_v(c)| \leq 2\epsilon$.

Proof. For any dimension d and group pair (g_i, g_j), define the true disparity as $\Delta_d(c; g_i, g_j) = |b_d(y_{c,g_i}) - b_d(y_{c,g_j})|$ and the estimated disparity as $\hat{\Delta}_d(c; g_i, g_j) = |\hat{b}_d(y_{c,g_i}) - \hat{b}_d(y_{c,g_j})|$. By the reverse triangle inequality,

$$\begin{aligned}\hat{\Delta}_d(c; g_i, g_j) - \Delta_d(c; g_i, g_j) \leq\ & |\hat{b}_d(y_{c,g_i}) - b_d(y_{c,g_i})| \\ & + |\hat{b}_d(y_{c,g_j}) - b_d(y_{c,g_j})|\end{aligned}. \quad (13)$$

Using the bounded-error assumption, we then obtain $|\hat{\Delta}_d(c; g_i, g_j) - \Delta_d(c; g_i, g_j)| \leq 2\epsilon_d$. Now consider the difference between the estimated and true fairness violation scores:

$$\begin{aligned}|\hat{F}_v(c) - F_v(c)| = |\tfrac{1}{|\mathcal{P}(c)|} \textstyle\sum_{(g_i,g_j)\in\mathcal{P}(c)} \sum_{d=1}^{K} \\ \omega_d \cdot (\hat{\Delta}_d(c; g_i, g_j) - \Delta_d(c; g_i, g_j))|\end{aligned}. \quad (14)$$

Applying the triangle inequality gives

$$\begin{aligned}|\hat{F}_v(c) - F_v(c)| \leq\ & \tfrac{1}{|\mathcal{P}(c)|} \textstyle\sum_{(g_i,g_j)\in\mathcal{P}(c)} \sum_{d=1}^{K} \\ & \omega_d \cdot |\hat{\Delta}_d(c; g_i, g_j) - \Delta_d(c; g_i, g_j)|\end{aligned}. \quad (15)$$

Substituting the bound $2\epsilon_d$, we obtain:

$$|\hat{F}_v(c) - F_v(c)| \leq \frac{1}{|\mathcal{P}(c)|} \sum_{(g_i,g_j)\in\mathcal{P}(c)} \sum_{d=1}^{K} \omega_d \cdot 2\epsilon_d. \quad (16)$$

Since the inner bound does not depend on the group pair, averaging over $|\mathcal{P}(c)|$ preserves it, yielding

$$|\hat{F}_v(c) - F_v(c)| \leq 2\sum_{d=1}^{K} \omega_d \cdot \epsilon_d. \quad (17)$$

If $\epsilon_d = \epsilon$, then because $\sum_{d=1}^{K} \omega_d = 1$, we have

$$|\hat{F}_v(c) - F_v(c)| \leq 2\epsilon. \quad (18)$$

This completes the proof.

Theorem 1 provides a robustness guarantee for BAIFT. Even if the individual behavioral scorers are imperfect, the overall fairness violation score remains stable as long as the dimension-level scoring errors are bounded. It is particularly important for BAIFT, since the framework combines rule-based, learned, and judge-based evaluators rather than relying on any single scoring source.

VII. Algorithm of BAIFT

Since fairness testing of LLMs can be expensive, BAIFT includes explicit budget control. The workflow can be bounded by: the maximum number of queried prompts, the maximum number of search iterations, the maximum number of retained cases, or the maximum number of calls to the generator or judge model. BAIFT stops when one of the following conditions is satisfied: the query budget is exhausted; the top-ranked fairness violation score has not improved for a preset patience window; or the coverage gain remains below a predefined threshold for the same patience window. This design allows BAIFT to trade off fairness coverage against computational cost.

In our implementation, BAIFT is organized as a self-contained toolkit with four modules: initializer, generator, evaluator, and interpreter. The toolkit supports both local and API-based target LLMs through a unified model client interface. Mutation operators are implemented as independent functions, making it easy to enable or disable specific search strategies. Behavioral scorers and diagnostic modules are similarly modular, allowing different evaluation backends to be plugged into the framework.

This modular design has two advantages. First, it improves reproducibility by clearly separating data preparation, test generation, evaluation, and interpretation. Second, it allows BAIFT to be extended in future work, for example by adding new behavioral dimensions, new mutation operators, or new diagnosis modules. The details of BAIFT are shown in **Algorithm 1**.

In practice, the candidate pool can be managed through beam search or an archive-based strategy. Beam search is suitable when the evaluation budget is limited, while archive-based search is suitable when broader exploration is desired.

Algorithm 1: BAIFT.

Input: target LLM f; initial task anchors $\mathcal{C}_0$; sensitive attributes $\mathcal{A}$; group generator $\mathcal{G}$; mutation operators $\mathbf{M}$; behavioral evaluator E; iteration budget T; beam size k.

Output: a ranked archive of fairness cases $\mathcal{R}$ with diagnostic reports.

Initialize candidate pool $\mathcal{Q} \leftarrow \mathcal{C}_0$, archive $\mathcal{R} \leftarrow \varnothing$, and iteration counter.
for *the iteration budget is not exhausted* **do**
  Select a batch of anchors from $\mathcal{Q}$.
  Apply constrained mutation operators to obtain new anchors.
  For each anchor, generate single-attribute and intersectional group configurations.
  Render prompts $p_{c,g}$ and query the target LLM to obtain responses $y_{c,g}$.
  Use E to compute behavioral scores, fairness violation score $F_v(c)$, amplification score $Amp(c)$, coverage gain $Cov(c)$, novelty $Nov(c)$, and invalidity penalty $Pen(c)$.
  Compute the prioritization score $J(c)$.
  Keep the top-k candidates according to $J(c)$ and update $\mathcal{Q}$.
  Add high-risk cases to $\mathcal{R}$.
end
Perform diagnostic labeling and visualization on $\mathcal{R}$.
Return $\mathcal{R}$.

A. System Components

BAIFT consists of four executable components:

- Initializer, which loads or creates task anchors and sensitive-attribute specifications;
- Generator, which renders anchors into group-specific prompts and applies constrained mutation operators;
- Evaluator, which queries the target LLM and computes behavioral fairness violation scores;
- Interpreter, which performs calibration, diagnostic labeling, and visualization.

Each component has clearly defined input and output interfaces. In implementation, task anchors, group configurations, prompts, responses, and fairness results are all stored in structured formats (e.g., JSON objects), so that the entire workflow can be replayed and audited.

B. Model Roles

BAIFT involves up to three different model roles.

Target LLM. This is the model under fairness testing. It receives group-specific prompts and produces the responses to be evaluated. The target LLM can be either a black-box API model or an open-source model deployed locally.

Generator model. This model is used to perform constrained paraphrasing or context expansion during case mutation. It is not the object of evaluation. Its role is limited

to generating alternative task anchors while preserving the original intent.

Judge model. This model is used only for part of the behavioral scoring process, especially for dimensions such as supportiveness or respectfulness that are difficult to capture with simple rules. To reduce evaluator contamination, the judge model is separated from the target LLM whenever possible. This separation is important in practice. It ensures that BAIFT does not rely on the target model to judge its own fairness behavior.

C. Data Structures

In implementation, BAIFT uses four core data structures. A task anchor is represented as a structured object containing: a case identifier, task intent, task context, response-format requirement, sensitive slots, and metadata such as domain and source.

A group configuration is a dictionary that assigns values to one or more sensitive attributes, e.g., {gender: female, age: older}.

A test case consists of a task anchor and a group configuration. A rendered test case is converted into a concrete prompt through the rendering function $p_{c,g} = \mathrm{Render}(c, g)$.

A fairness record stores: the rendered prompts, the generated responses, the behavioral scores, the fairness violation score $F_v(c)$, the amplification score $Amp(c)$, the diagnostic labels, and the visualization metadata. These structures allow BAIFT to support both online evaluation and offline analysis.

D. End-to-end Workflow

The end-to-end workflow of BAIFT is shown in **Algorithm 1**. It proceeds in five stages.

Stage 1: **Anchor initialization**. BAIFT begins with a set of seed task anchors $\mathcal{C}_0$. These anchors may be manually authored, collected from task templates, or produced through constrained paraphrasing.

Stage 2: **Candidate generation**. For each candidate anchor, BAIFT applies one or more mutation operators to produce new variants. Each anchor is then rendered into prompts for both single-attribute and intersectional group configurations.

Stage 3: **Target-model querying**. All rendered prompts are sent to the target LLM. The resulting responses are stored together with their corresponding case and group identifiers.

Stage 4: **Behavioral fairness evaluation**. The evaluator computes the behavioral profile $B(y)$ for each response, then derives the fairness violation score $F_v(c)$, the amplification score $Amp(c)$, and the prioritization score $J(c)$.

Stage 5: **Prioritization and interpretation**. BAIFT ranks candidate cases according to $J(c)$, updates the archive of selected cases, and produces diagnostic labels and visual summaries for high-risk cases. The process iterates until a predefined budget or stopping criterion is reached.

E. Complexity Discussion of BAIFT

The computational cost of BAIFT is mainly determined by the number of candidate anchors, the number of group configurations per anchor, and the cost of querying the target LLM. Let $|\mathcal{Q}|$ be the number of candidate anchors in one iteration and let $|\mathcal{G}_c|$ be the number of rendered group configurations per case. The dominant cost of one iteration is:

$$O\left(\sum_{c\in\mathcal{Q}} |\mathcal{G}_c| \cdot Cost(f)\right) \tag{19}$$

where $Cost(f)$ denotes the cost of one target-model query. The behavioral evaluation and diagnostic analysis are comparatively lightweight, except when an external judge model is used. The prioritization strategy helps BAIFT reduce unnecessary exploration by focusing the search on cases with high fairness risk or high coverage gain.

TABLE V
HUMAN ANNOTATION PROTOCOL USED FOR CALIBRATION AND RQ4 VALIDATION.

Annotation Target	Annotator Task	Aggregation	Used For
Unfairness severity	Rate whether one group receives less favorable treatment on a five-point Likert scale.	Average score over three annotators.	Spearman correlation with $F_v(c)$.
Affected HIS2Rs dimensions	Mark dimensions where assistance, information, support, respect, or refusal differs across groups.	Multi-label majority vote.	Fleiss' Kappa and $\mathbf{\Delta}(c)$ validation.
Vulnerable group	Identify the group that receives less favorable behavior in the paired responses.	Majority vote; ties are treated as uncertain labels.	Accuracy of $VG(g)$-based ranking.
Intersectional amplification	Decide whether the joint group condition appears worse than the corresponding single-attribute condition.	Binary majority vote.	F1 score of $Amp(c) > 0$ detection.

VIII. IMPLEMENTATION DETAILS

To ensure a fair and reproducible comparison, we use the same initial task anchors, sensitive attributes, target models, and query budgets for BAIFT and all baselines. Each method is first executed with its original test-generation strategy to produce candidate test cases. For methods that do not natively support intersectional testing, we extend them with the same group-rendering mechanism used in BAIFT, while keeping their original generation and selection logic unchanged. This allows all methods to be evaluated under the same sensitive-attribute space.

Unified violation judgment. Since different methods adopt different fairness oracles, we decouple test-case generation from violation judgment. All generated cases are re-evaluated using the same post-hoc criteria. For explicit discriminatory instances, we follow the semantic-consistency criterion used in CAFFE and compute the LSA-based similarity score $Sim(c)$ between counterfactual responses. A case is counted as an explicit discriminatory instance if $Sim(c) < 0.9$. For implicit discriminatory instances, we apply the unified behavior-aware scorers to compute $F_v(c)$. The threshold τ is determined from a shared calibration set using the 3σ rule. A case is counted as an implicit discriminatory instance if $Sim(c) \geq 0.9$ and $F_v(c) > \tau$. Thus, all methods are compared under the same explicit and implicit violation criteria.

Calibration of behavior-aware scorers. To avoid favoring BAIFT, the behavior-aware scorers are calibrated on a shared set of cases uniformly sampled from the outputs of all methods. This calibration set is used to estimate the threshold τ for $F_v(c)$ and to tune the weights of the hybrid scorers. After calibration, the same scorers and thresholds are fixed and applied to all methods, models, and experimental settings.

Intersectional attribute selection. We consider 11 sensitive attributes in total. Exhaustively testing all pairwise combinations would produce $C_{11}^2 = 55$ attribute pairs, and higher-order combinations would further increase the search space. Therefore, we adopt a two-stage strategy. We first conduct a lightweight pilot screening over all 55 pairs using a small set of anchors and representative models. For each pair (A_i, A_j), we compute a preliminary risk score based on the average behavioral violation score and average amplification score. We then select five high-risk and semantically diverse attribute pairs for full evaluation. This strategy enlarges intersectional coverage while keeping the experiment tractable.

Coverage and fine-tuning evaluation. For coverage analysis, we randomly sample the same number of cases from each method and compute thresholded IoU curves over multiple similarity thresholds. Each sampling process is repeated five times, and we report the mean and variance as well as the AUC of the IoU curve. For fairness-guided fine-tuning, we sample the same number of generated cases from each method to fine-tune the same base model. The fine-tuned models are evaluated on held-out cases from other methods using $Sim(c)$ and $F_v(c)$.

Efficiency and generalizability settings. For efficiency evaluation, all methods are run under the same query budget. We report both time-based and query-normalized metrics, including Time/EDIs, Time/IDIs, #EDIs per 1K queries, and #IDIs per 1K queries. For generalizability evaluation, we aggregate results across model scale, architecture, modality, sensitive attributes, intents, and deployment settings. API-based and local models are evaluated separately when reporting efficiency.

Human annotation and statistical testing. For diagnostic reliability, we randomly sample test cases from different methods and ask annotators to evaluate unfairness severity, affected behavioral dimensions, vulnerable groups, and intersectional amplification. The annotation results are used only for calibration and reliability analysis, not for generating test cases. We compute agreement between BAIFT diagnostics and human annotations using correlation, Fleiss' Kappa, Accuracy, and F1 score. Unless otherwise specified, all randomized experiments are repeated five times. We report average results and use statistical tests to assess whether the differences between BAIFT and baselines are significant.

Table V details the annotation protocol. Each response pair is labeled independently by three annotators. For scalar unfairness severity, we use the averaged Likert score as the human judgment. For affected dimensions, vulnerable groups, and amplification labels, we use majority voting and record disagreements for reliability analysis.

The first ten models in Table VI are used as the main target LLMs in RQ1–RQ4. Qwen3-8B is used only as the base model for the fairness-guided fine-tuning experiment in RQ1.

TABLE VI
TARGET LLMS AND THE FINE-TUNING BASE MODEL USED IN THE EXPERIMENTS.

LLMs	Date	Architecture	Size	Open Source	Vendors
GPT-5.4	2026-03	-	-	No	OpenAI
Gemini-3-Flash	2025-12	-	-	No	Google
Llama-3.1-8B	2024-07	Dense	8B	Yes	Meta
Llama-3.3-70B	2024-12	Dense	70B	Yes	Meta
Mixtral-8×7B	2023-12	MoE	46.7B	Yes	Mistral AI
DeepSeek-V3.2	2025-12	MoE	685B	Yes	DeepSeek
Kimi-K2.5	2026-01	MoE	1T	Yes	Moonshot
GLM-5	2026-02	MoE	744B	Yes	Z.ai
Qwen3-14B	2025-04	Dense	14B	Yes	Alibaba
Qwen3-235B	2025-07	MoE	235B	Yes	Alibaba
Qwen3-8B	2025-04	Dense	8B	Yes	Alibaba

TABLE VII
CORRESPONDENCE BETWEEN THE MAIN-PAPER SUBJECT CONFIGURATIONS AND SUPPLEMENTARY RECORDS.

Main Subject IDs	RQ/Purpose	Scope	Supplementary Evidence	Main Metrics
1–22	RQ1/RQ3	Single-attribute testing over 10 target LLMs and 11 sensitive attributes	Raw single-attribute effectiveness records and model-level generalizability records	#DIs, #IDIs, $F_v(c)$
23–32	RQ1/RQ3	Intersectional testing over five screened attribute pairs and 10 target LLMs	Raw pair-level and model-level intersectional records	#EDIs, #IDIs, IDI Ratio, $\overline{Amp(c)}$, AmpRatio
33–34	RQ1	Fairness-guided fine-tuning on Qwen3-8B	Fine-tuning guidance table with held-out evaluation	$Sim(c)$, $F_v(c)$
35–38	RQ4	Human annotation over four task intents	Annotation protocol, calibration process, and diagnosis reliability records	Spearman, Fleiss' Kappa, Accuracy, F1
39–44	RQ5	Component ablation on GPT-5.4, DeepSeek-V3.2, and Qwen3-14B	Raw and averaged ablation tables	#EDIs, #IDIs, IDI Ratio, $\overline{F_v(c)}$, $\overline{Amp(c)}$, Time/EDIs, Time/IDIs
45	RQ1	Pilot screening over all 55 candidate attribute pairs	Intersectional attribute-selection protocol and selected five high-risk pairs	Preliminary $F_v(c)$ and $Amp(c)$ risk
46	Setup/RQ4	Threshold calibration and reliability checking	Shared scorer calibration and unified post-hoc judgment protocol	τ, $F_v(c)$, human agreement metrics

TABLE VIII
RQ1 SINGLE-ATTRIBUTE EFFECTIVENESS RESULTS.

Target LLMs	METAL		CAFFE		LLM-Roleplay		BAIFT	
	#DIs	#IDIs	#DIs	#IDIs	#DIs	#IDIs	#DIs	#IDIs
GPT-5.4	362	105	834	232	614	128	956	584
Gemini-3-Flash	548	167	1,187	243	894	173	1,798	812
Llama-3.1-8B	2,920	505	3,645	998	3,584	895	4,461	3,042
Llama-3.3-70B	1,627	495	3,034	786	2,248	508	3,708	1,995
Mixtral-8×7B	2,391	546	3,826	961	3,127	732	4,413	2,635
DeepSeek-V3.2	731	246	1,635	419	1,168	312	1,994	1,037
Kimi-K2.5	1,138	348	2,269	668	1,673	338	2,907	1,514
GLM-5	1,362	458	2,628	779	1,936	452	3,801	1,742
Qwen3-14B	1,974	652	3,457	978	2,689	612	3,926	2,287
Qwen3-235B	914	262	1,886	534	1,397	351	2,578	1,266
Average	1,397	378	2,440	660	1,933	450	3,054	1,691
p-value	1.84E-6	6.16E-5	3.16E-5	8.31E-5	6.12E-6	3.32E-5	–	–

IX. EXPERIMENTAL RAW DATA RECORDS

This section records the experimental statistics used in the main paper. Unless otherwise specified, all ratios and scores are stored as numeric values in the range $[0, 1]$. The main manuscript reports compact tables and figures, while this section preserves the source statistics used to generate them.

A. RQ1: Effectiveness and Coverage

Table VIII reports the model-level single-attribute results used to support RQ1, including the total detected discriminatory instances and the subset of implicit discriminatory instances for each method.

Table IX summarizes the intersectional results over the five selected high-risk attribute pairs, showing how each method performs on explicit violations, implicit violations, implicit ratio, and amplification-related metrics.

Table X gives the model-level raw results for the <Gender, Age> pair, which is used to examine whether gender- and age-related disparities are amplified when the two attributes are tested jointly.

Table XI reports the same intersectional statistics for the <Race, Religion> pair, enabling a direct comparison of racial and religious bias signals across target LLMs and testing methods.

Table XII records the raw results for the <Social, Occupation> pair, which captures whether socioeconomic status and occupational identity jointly increase behavioral unfairness.

TABLE IX
RQ1 INTERSECTIONAL EFFECTIVENESS SUMMARY.

Attribute Pairs	METAL					CAFFE					LLM-Roleplay					BAIFT				
	#EDIs	#IDIs	Ratio	$\overline{Amp(c)}$	AmpRatio	#EDIs	#IDIs	Ratio	$\overline{Amp(c)}$	AmpRatio	#EDIs	#IDIs	Ratio	$\overline{Amp(c)}$	AmpRatio	#EDIs	#IDIs	Ratio	$\overline{Amp(c)}$	AmpRatio
<Gender, Age>	389	142	0.27	0.08	0.23	619	307	0.33	0.10	0.37	501	293	0.37	0.01	0.30	661	807	0.55	0.13	0.58
<Race, Religion>	424	236	0.36	0.03	0.26	673	355	0.35	-0.01	0.41	541	311	0.37	0.01	0.34	741	783	0.51	0.04	0.62
<Social, Occupation>	395	204	0.34	0.00	0.25	640	343	0.35	0.05	0.40	512	332	0.39	0.05	0.33	680	771	0.53	0.03	0.61
<Ability, Body>	418	228	0.35	0.01	0.26	659	330	0.33	0.00	0.41	532	293	0.36	-0.11	0.34	696	770	0.53	0.07	0.63
<Culture, Victim>	391	173	0.31	-0.03	0.24	632	343	0.35	0.02	0.39	501	300	0.37	-0.01	0.32	707	729	0.51	0.05	0.60
Average	403	197	0.33	0.02	0.25	645	336	0.34	0.03	0.40	517	306	0.37	-0.01	0.33	697	772	0.53	0.06	0.61
p-value	3.37E-6	7.32E-6	3.91E-4	8.91E-3	4.24E-9	1.35E-3	9.28E-6	3.32E-5	4.18E-2	4.49E-8	2.49E-5	3.28E-6	3.71E-5	4.90E-2	4.70E-9	–	–	–	–	–

TABLE X
RQ1 INTERSECTIONAL RAW RESULTS FOR <GENDER, AGE>.

Target LLM	METAL					CAFFE					LLM-Roleplay					BAIFT				
	EDI	IDI	Ratio	Amp	AmpR	EDI	IDI	Ratio	Amp	AmpR	EDI	IDI	Ratio	Amp	AmpR	EDI	IDI	Ratio	Amp	AmpR
GPT-5.4	12	7	0.37	0.16	0.06	27	14	0.34	-0.08	0.14	18	10	0.36	0.21	0.10	34	43	0.56	-0.14	0.26
Gemini-3-Flash	18	11	0.38	-0.27	0.09	39	23	0.37	0.34	0.22	25	16	0.39	0.05	0.14	45	56	0.55	0.42	0.35
Llama-3.1-8B	68	12	0.15	-0.09	0.42	84	58	0.41	0.31	0.56	79	53	0.40	-0.73	0.50	78	91	0.54	0.74	0.82
Llama-3.3-70B	47	29	0.38	0.23	0.28	74	18	0.20	-0.63	0.47	61	39	0.39	0.11	0.37	78	99	0.56	0.58	0.70
Mixtral-8×7B	61	8	0.12	0.66	0.37	81	35	0.30	-0.17	0.52	73	29	0.28	0.49	0.46	85	96	0.53	-0.61	0.77
DeepSeek-V3.2	24	15	0.38	0.48	0.13	48	29	0.38	-0.31	0.27	34	21	0.38	-0.19	0.19	54	68	0.56	0.17	0.46
Kimi-K2.5	35	8	0.19	0.71	0.21	63	24	0.28	0.08	0.37	48	30	0.38	-0.52	0.26	69	88	0.56	0.29	0.60
GLM-5	41	25	0.38	-0.39	0.25	69	21	0.23	0.46	0.41	55	35	0.39	0.64	0.32	74	94	0.56	-0.07	0.64
Qwen3-14B	54	9	0.14	-0.55	0.32	78	51	0.40	0.41	0.49	67	34	0.34	-0.28	0.42	82	93	0.53	0.36	0.73
Qwen3-235B	29	18	0.38	-0.12	0.16	56	34	0.38	0.59	0.32	41	26	0.39	0.33	0.24	62	79	0.56	-0.46	0.52

TABLE XI
RQ1 INTERSECTIONAL RAW RESULTS FOR <RACE, RELIGION>.

Target LLM	METAL					CAFFE					LLM-Roleplay					BAIFT				
	EDI	IDI	Ratio	Amp	AmpR	EDI	IDI	Ratio	Amp	AmpR	EDI	IDI	Ratio	Amp	AmpR	EDI	IDI	Ratio	Amp	AmpR
GPT-5.4	14	8	0.36	0.27	0.07	31	16	0.34	-0.34	0.18	21	12	0.36	0.18	0.12	39	48	0.55	-0.22	0.31
Gemini-3-Flash	20	13	0.39	-0.42	0.11	43	25	0.37	0.51	0.26	29	18	0.38	-0.16	0.17	52	63	0.55	0.39	0.39
Llama-3.1-8B	72	18	0.20	0.19	0.46	93	63	0.40	-0.47	0.61	81	28	0.26	-0.25	0.53	96	91	0.49	0.71	0.85
Llama-3.3-70B	52	32	0.38	0.36	0.33	79	51	0.39	-0.72	0.49	67	45	0.40	-0.54	0.42	86	103	0.54	0.28	0.73
Mixtral-8×7B	63	39	0.38	0.74	0.42	88	28	0.24	-0.26	0.54	74	31	0.30	-0.39	0.47	91	32	0.26	0.62	0.77
DeepSeek-V3.2	27	17	0.39	0.58	0.16	53	31	0.37	-0.29	0.32	38	24	0.39	0.44	0.25	61	74	0.55	-0.65	0.48
Kimi-K2.5	39	24	0.38	-0.61	0.24	68	22	0.24	0.43	0.41	54	35	0.39	0.57	0.34	77	94	0.55	-0.48	0.63
GLM-5	46	28	0.38	-0.18	0.29	74	48	0.39	0.69	0.46	61	41	0.40	0.31	0.38	82	99	0.55	-0.07	0.70
Qwen3-14B	58	36	0.38	-0.53	0.38	83	34	0.29	0.14	0.52	70	48	0.41	0.66	0.44	88	95	0.52	-0.33	0.76
Qwen3-235B	33	21	0.39	-0.09	0.20	61	37	0.38	0.22	0.36	46	29	0.39	-0.71	0.29	69	84	0.55	0.17	0.57

TABLE XII
RQ1 INTERSECTIONAL RAW RESULTS FOR <SOCIAL, OCCUPATION>.

Target LLM	METAL					CAFFE					LLM-Roleplay					BAIFT				
	EDI	IDI	Ratio	Amp	AmpR	EDI	IDI	Ratio	Amp	AmpR	EDI	IDI	Ratio	Amp	AmpR	EDI	IDI	Ratio	Amp	AmpR
GPT-5.4	11	6	0.35	0.31	0.06	26	13	0.33	-0.27	0.15	18	9	0.33	0.14	0.11	34	42	0.55	-0.36	0.28
Gemini-3-Flash	17	10	0.37	-0.44	0.10	37	21	0.36	0.52	0.23	25	15	0.38	-0.18	0.16	45	54	0.55	0.28	0.35
Llama-3.1-8B	69	16	0.19	-0.15	0.44	89	30	0.25	0.29	0.59	80	54	0.40	-0.31	0.52	95	84	0.47	0.76	0.88
Llama-3.3-70B	49	31	0.39	0.24	0.31	77	49	0.39	-0.33	0.48	63	22	0.26	0.73	0.41	81	99	0.55	-0.12	0.73
Mixtral-8×7B	62	38	0.38	0.27	0.40	86	28	0.25	-0.20	0.55	75	50	0.40	0.55	0.48	81	83	0.51	-0.67	0.83
DeepSeek-V3.2	23	14	0.38	0.63	0.14	47	28	0.37	-0.12	0.28	33	21	0.39	0.47	0.21	55	67	0.55	-0.51	0.44
Kimi-K2.5	36	23	0.39	0.49	0.23	66	41	0.38	-0.58	0.40	50	33	0.40	0.22	0.32	72	88	0.55	-0.24	0.61
GLM-5	43	27	0.39	-0.71	0.27	72	45	0.38	0.16	0.44	57	37	0.39	-0.09	0.35	77	94	0.55	0.61	0.67
Qwen3-14B	55	20	0.27	-0.56	0.35	82	53	0.39	0.64	0.51	69	64	0.48	-0.41	0.45	76	81	0.52	0.38	0.78
Qwen3-235B	30	19	0.39	-0.08	0.18	58	35	0.38	0.35	0.34	42	27	0.39	-0.66	0.27	64	79	0.55	0.19	0.53

Table XIII presents the raw results for the <Ability, Body> pair, supporting the analysis of intersectional risks involving disability-related and physical-appearance-related attributes.

Table XIV reports the raw results for the <Culture, Victim> pair, showing whether cultural background and victim status jointly affect the detected fairness violations.

Table XV provides the thresholded IoU curves and AUC values used for the coverage analysis, where higher overlap with baseline regions indicates broader coverage of behaviorally relevant test areas.

Table XVI records the held-out evaluation after using the generated cases for fairness-guided fine-tuning, comparing whether cases from different methods can improve semantic consistency and reduce behavioral violation scores.

B. RQ2: Efficiency

Table XVII reports the efficiency measurements under API-based and local deployments, including the time cost per detected explicit or implicit violation and the number of violations detected per 1K queries.

TABLE XIII
RQ1 INTERSECTIONAL RAW RESULTS FOR <ABILITY, BODY>.

Target LLM	METAL					CAFFE					LLM-Roleplay					BAIFT				
	EDI	IDI	Ratio	Amp	AmpR	EDI	IDI	Ratio	Amp	AmpR	EDI	IDI	Ratio	Amp	AmpR	EDI	IDI	Ratio	Amp	AmpR
GPT-5.4	13	7	0.35	-0.22	0.07	29	15	0.34	0.38	0.17	20	11	0.35	-0.47	0.13	37	46	0.55	0.31	0.30
Gemini-3-Flash	19	12	0.39	0.55	0.12	40	23	0.37	-0.15	0.25	28	17	0.38	0.26	0.18	48	58	0.55	-0.39	0.38
Llama-3.1-8B	73	45	0.38	0.21	0.47	91	26	0.22	-0.68	0.62	84	34	0.29	-0.36	0.56	94	85	0.47	0.70	0.91
Llama-3.3-70B	51	32	0.39	-0.11	0.33	78	50	0.39	0.77	0.48	64	43	0.40	0.58	0.41	73	76	0.51	-0.08	0.74
Mixtral-8×7B	64	9	0.12	-0.59	0.41	88	28	0.24	0.33	0.56	76	16	0.17	0.17	0.51	80	73	0.48	-0.45	0.84
DeepSeek-V3.2	26	16	0.38	-0.67	0.15	51	30	0.37	0.49	0.31	36	23	0.39	-0.05	0.24	58	70	0.55	0.64	0.47
Kimi-K2.5	38	24	0.39	-0.34	0.25	67	42	0.39	0.18	0.42	52	34	0.40	0.41	0.33	74	90	0.55	-0.56	0.63
GLM-5	45	28	0.38	0.69	0.28	73	46	0.39	-0.41	0.45	58	39	0.40	-0.29	0.37	79	96	0.55	0.22	0.69
Qwen3-14B	57	35	0.38	0.44	0.37	83	34	0.29	-0.24	0.52	70	47	0.40	-0.62	0.46	87	94	0.52	0.53	0.79
Qwen3-235B	32	20	0.38	0.12	0.21	59	36	0.38	-0.62	0.36	44	29	0.40	-0.73	0.29	66	82	0.55	-0.18	0.56

TABLE XIV
RQ1 INTERSECTIONAL RAW RESULTS FOR <CULTURE, VICTIM>.

Target LLM	METAL					CAFFE					LLM-Roleplay					BAIFT				
	EDI	IDI	Ratio	Amp	AmpR	EDI	IDI	Ratio	Amp	AmpR	EDI	IDI	Ratio	Amp	AmpR	EDI	IDI	Ratio	Amp	AmpR
GPT-5.4	10	5	0.33	0.46	0.05	24	12	0.33	-0.19	0.14	16	8	0.33	0.33	0.09	32	40	0.56	-0.27	0.26
Gemini-3-Flash	16	9	0.36	-0.32	0.09	35	20	0.36	0.61	0.22	24	14	0.37	-0.11	0.14	43	52	0.55	0.49	0.34
Llama-3.1-8B	71	13	0.15	-0.28	0.45	93	33	0.26	0.12	0.61	83	27	0.25	-0.16	0.54	96	73	0.43	0.74	0.90
Llama-3.3-70B	48	30	0.38	0.67	0.30	76	48	0.39	-0.22	0.47	62	41	0.40	0.24	0.40	80	98	0.55	-0.09	0.72
Mixtral-8×7B	63	5	0.07	0.09	0.41	87	35	0.29	-0.31	0.55	75	51	0.40	0.52	0.49	90	86	0.49	-0.57	0.84
DeepSeek-V3.2	22	13	0.37	0.18	0.12	46	27	0.37	-0.54	0.28	31	20	0.39	0.31	0.20	53	65	0.55	-0.63	0.43
Kimi-K2.5	35	21	0.38	0.29	0.22	64	39	0.38	-0.46	0.40	48	31	0.39	0.18	0.31	70	86	0.55	-0.38	0.59
GLM-5	42	26	0.38	-0.14	0.26	70	44	0.39	0.35	0.43	55	36	0.40	-0.69	0.36	76	81	0.52	0.66	0.65
Qwen3-14B	56	34	0.38	-0.51	0.35	82	52	0.39	0.78	0.51	68	46	0.40	-0.44	0.45	105	71	0.40	0.37	0.77
Qwen3-235B	28	17	0.38	-0.76	0.17	55	33	0.38	0.07	0.34	39	26	0.40	-0.25	0.25	62	77	0.55	0.15	0.51

TABLE XV
RQ1 COVERAGE STATISTICS OVER DIFFERENT τ THRESHOLDS.

τ	IoU(METAL, CAFFE)	IoU(METAL, LLM-Roleplay)	IoU(CAFFE, LLM-Roleplay)	IoU(METAL, BAIFT)	IoU(CAFFE, BAIFT)	IoU(LLM-Roleplay, BAIFT)
0.00	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
0.05	0.1891	0.1669	0.2063	0.5662	0.6835	0.3459
0.10	0.1772	0.1370	0.1947	0.5291	0.6327	0.3219
0.15	0.1655	0.1370	0.1820	0.4826	0.5760	0.2747
0.20	0.1422	0.1370	0.1461	0.4472	0.5326	0.2747
0.25	0.1422	0.1274	0.1461	0.4025	0.4738	0.2555
0.30	0.1312	0.1167	0.1461	0.3012	0.4296	0.2129
0.35	0.1211	0.1074	0.1344	0.3012	0.3822	0.2129
0.40	0.1105	0.0995	0.1242	0.3012	0.2723	0.1947
0.45	0.1007	0.0905	0.1130	0.2666	0.2723	0.1744
0.50	0.0911	0.0817	0.1019	0.2392	0.2392	0.1567
0.55	0.0817	0.0741	0.0923	0.2099	0.2392	0.1403
0.60	0.0730	0.0655	0.0817	0.1841	0.2077	0.1242
0.65	0.0644	0.0576	0.0718	0.1587	0.1779	0.1086
0.70	0.0560	0.0504	0.0627	0.1370	0.1534	0.0923
0.75	0.0477	0.0428	0.0532	0.1148	0.1287	0.0787
0.80	0.0230	0.0357	0.0444	0.0947	0.1056	0.0373
0.85	0.0230	0.0283	0.0352	0.0747	0.0828	0.0373
0.90	0.0230	0.0209	0.0157	0.0331	0.0599	0.0373
0.95	0.0142	0.0127	0.0157	0.0331	0.0363	0.0230
1.00	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
AUC	0.1388	0.1295	0.1484	0.2939	0.3343	0.2052

TABLE XVI
RQ1 FINE-TUNING GUIDANCE RESULTS.

Metric	Target LLMs	METAL	CAFFE	LLM-Roleplay	BAIFT
Sim(c)	Qwen3-8B	0.41	0.38	0.38	0.19
Sim(c)	Qwen3-8B-FT-METAL	0.66	0.42	0.48	0.31
Sim(c)	Qwen3-8B-FT-CAFFE	0.89	0.91	0.92	0.86
Sim(c)	Qwen3-8B-FT-LLM-Roleplay	0.70	0.68	0.71	0.54
Sim(c)	Qwen3-8B-FT-BAIFT	0.93	0.91	0.94	0.93
$F_v(c)$	Qwen3-8B	0.57	0.61	0.63	0.79
$F_v(c)$	Qwen3-8B-FT-METAL	0.35	0.54	0.53	0.68
$F_v(c)$	Qwen3-8B-FT-CAFFE	0.25	0.19	0.23	0.36
$F_v(c)$	Qwen3-8B-FT-LLM-Roleplay	0.31	0.29	0.30	0.45
$F_v(c)$	Qwen3-8B-FT-BAIFT	0.09	0.13	0.12	0.10

TABLE XVII
RQ2 EFFICIENCY RESULTS.

Scenario	Method	Time/EDIs	Time/IDIs	#EDIs/1K Queries	#IDIs/1K Queries
API calls	METAL	90.91	250.00	11.00	4.00
API calls	CAFFE	55.55	166.67	18.00	6.00
API calls	LLM-Roleplay	66.67	200.00	15.00	5.00
API calls	BAIFT	71.43	58.82	14.00	17.00
Local calls	METAL	79.17	190.00	12.00	5.00
Local calls	CAFFE	50.00	158.33	19.00	6.00
Local calls	LLM-Roleplay	67.86	237.50	14.00	4.00
Local calls	BAIFT	63.33	52.78	15.00	18.00

TABLE XVIII
RQ3 GENERALIZABILITY ACROSS TARGET LLMS.

Target LLMs	METAL		CAFFE		LLM-Roleplay		BAIFT	
	#DIs	$F_v(c)$	#DIs	$F_v(c)$	#DIs	$F_v(c)$	#DIs	$F_v(c)$
GPT-5.4	362	0.26	834	0.32	614	0.35	956	0.55
Gemini-3-Flash	548	0.31	1,187	0.35	894	0.41	1,798	0.56
Llama-3.1-8B	2,920	0.48	3,645	0.56	3,584	0.59	4,461	0.75
Llama-3.3-70B	1,627	0.38	3,034	0.48	2,248	0.51	3,708	0.68
Mixtral-8×7B	2,391	0.42	3,826	0.52	3,127	0.52	4,413	0.72
DeepSeek-V3.2	731	0.29	1,635	0.39	1,168	0.40	1,994	0.61
Kimi-K2.5	1,138	0.31	2,269	0.43	1,673	0.43	2,907	0.62
GLM-5	1,362	0.35	2,628	0.42	1,936	0.46	3,801	0.58
Qwen3-14B	1,974	0.46	3,457	0.52	2,689	0.51	3,926	0.73
Qwen3-235B	914	0.32	1,886	0.41	1,397	0.42	2,578	0.64
Average	1,396.70	0.36	2,440.10	0.44	1,933.00	0.46	3,054.20	0.64

TABLE XIX
RQ3 GENERALIZABILITY ACROSS TASK INTENTS.

Intent	METAL		CAFFE		LLM-Roleplay		BAIFT	
	#DIs	$F_v(c)$	#DIs	$F_v(c)$	#DIs	$F_v(c)$	#DIs	$F_v(c)$
Question	3,859	0.32	6,485	0.43	5,685	0.47	8,457	0.66
Recommendation	2,685	0.39	7,564	0.51	4,524	0.42	7,156	0.71
Direction	4,591	0.32	6,428	0.46	3,650	0.38	5,283	0.65
Clarification	2,832	0.39	3,924	0.45	5,471	0.38	9,646	0.63
Average	3,491.75	0.36	6,100.25	0.46	4,832.50	0.41	7,635.50	0.66

C. RQ3: Generalizability

Table XVIII reports the generalizability results across target LLMs, showing whether each method can consistently detect discriminatory instances and high behavioral violation scores on different model families.

Table XIX aggregates the results by task intent, which is used to evaluate whether BAIFT remains effective across question answering, recommendation, direction, and clarification scenarios.

Table XX aggregates the results by sensitive attribute and selected intersectional attribute pair, supporting the claim that BAIFT generalizes across both single-attribute and intersectional fairness settings.

Table XXI groups the results by model architecture, modality, and deployment setting, providing the raw evidence for the six-dimensional generalizability analysis in the main paper.

D. RQ4: Diagnostic Reliability

Table XXII reports the reliability metrics used to validate BAIFT's diagnostic outputs against semantic disparity, HumanScore, annotator agreement, vulnerable-group labels, and amplification labels.

Table XXIII lists the average behavior-dimension diagnosis scores for each target LLM, showing which HIS2Rs dimensions contribute most to the detected fairness violations.

Table XXIV reports the top-ranked vulnerable groups for each target LLM according to $VG(g)$, providing group-level evidence for the diagnosis module.

Table XXV extends the main figure by reporting the top-five intersectional group pairs according to $Amp(c)$, showing which combinations exhibit the strongest amplification effects in the diagnosis results.

TABLE XX
RQ3 GENERALIZABILITY ACROSS SENSITIVE ATTRIBUTES.

Attribute	METAL		CAFFE		LLM-Roleplay		BAIFT	
	#DIs	$F_v(c)$	#DIs	$F_v(c)$	#DIs	$F_v(c)$	#DIs	$F_v(c)$
ability	1,585	0.32	2,055	0.44	1,543	0.37	886	0.61
age	1,462	0.38	2,165	0.46	524	0.46	3,591	0.64
body	1,754	0.31	1,944	0.42	2,356	0.51	2,652	0.69
character	1,646	0.42	4,153	0.53	2,514	0.51	2,515	0.71
culture	1,947	0.35	1,582	0.48	1,642	0.39	2,647	0.59
gender	1,053	0.38	2,317	0.45	953	0.51	2,933	0.64
occupation	453	0.29	1,275	0.38	1,440	0.46	3,561	0.61
race	1,685	0.45	1,352	0.49	1,641	0.48	3,644	0.58
religion	948	0.33	2,251	0.42	1,958	0.43	3,489	0.61
social	521	0.42	2,927	0.39	1,717	0.54	2,682	0.69
victim	913	0.38	2,380	0.48	3,042	0.47	1,942	0.67
<Gender, Age>	531	0.34	926	0.46	794	0.35	1,468	0.61
<Race, Religion>	660	0.42	1,028	0.51	852	0.34	1,524	0.72
<Social, Occupation>	599	0.29	983	0.46	844	0.46	1,451	0.54
<Ability, Body>	646	0.41	989	0.51	825	0.43	1,466	0.64
<Culture, Victim>	564	0.38	975	0.48	801	0.41	1,436	0.69
Average	1,060.44	0.37	1,831.38	0.46	1,465.38	0.45	2,367.94	0.64

TABLE XXI
RQ3 GENERALIZABILITY ACROSS MODEL ARCHITECTURES, MODALITIES, AND DEPLOYMENTS.

Dimension	Value	METAL		CAFFE		LLM-Roleplay		BAIFT	
		#DIs	F_v	#DIs	F_v	#DIs	F_v	#DIs	F_v
Architecture	Dense	2,173.67	0.44	3,378.67	0.52	2,840.33	0.54	4,031.67	0.72
Architecture	MoE	1,307.20	0.34	2,448.80	0.43	1,860.20	0.45	3,138.60	0.63
Architecture	Average	1,740.43	0.39	2,913.73	0.48	2,350.27	0.49	3,585.13	0.68
Modality	Text-only	1,702.71	0.39	2,873.00	0.47	2,307.00	0.49	3,554.43	0.67
Modality	Multimodal	682.67	0.29	1,430.00	0.37	1,060.33	0.40	1,887.00	0.58
Modality	Average	1,192.69	0.34	2,151.50	0.42	1,683.67	0.44	2,720.71	0.62
Deployment	API calls	1,134.13	0.33	2,162.38	0.42	1,632.13	0.44	2,769.38	0.62
Deployment	Local calls	2,447.00	0.47	3,551.00	0.54	3,136.50	0.55	4,193.50	0.74
Deployment	Average	1,790.56	0.40	2,856.69	0.48	2,384.31	0.49	3,481.44	0.68

TABLE XXII
RQ4 DIAGNOSTIC METRIC RELIABILITY.

Component	Metric	Value	Interpretation
Fairness Violation Score $F_v(c)$	Spearman($F_v(c)$, $1 - Sim(c)$)	0.68	Consistency with semantic disparity
Fairness Violation Score $F_v(c)$	Spearman($F_v(c)$, HumanScore)	0.72	Alignment with human judgment
Behavior Diagnosis $\mathbf{\Delta}(c)$	Average Fleiss' Kappa	0.67	Agreement on affected behaviors
Vulnerable Group $VG(g)$	Accuracy	0.87	Correct group identification
Amplification $Amp(c)$	F1 measure	0.82	Correct amplification detection

TABLE XXIII
RQ4 BEHAVIOR-DIMENSION DIAGNOSIS SCORES.

Target LLMs	Δ_H	Δ_I	Δ_S	Δ_R	Δ_{Rs}
GPT-5.4	0.34	0.47	0.40	0.35	0.52
Gemini-3-Flash	0.38	0.48	0.42	0.35	0.52
Llama-3.1-8B	0.58	0.69	0.62	0.51	0.66
Llama-3.3-70B	0.48	0.57	0.55	0.43	0.61
Mixtral-8×7B	0.48	0.64	0.61	0.50	0.77
DeepSeek-V3.2	0.43	0.51	0.47	0.40	0.60
Kimi-K2.5	0.45	0.56	0.50	0.39	0.62
GLM-5	0.45	0.52	0.45	0.38	0.58
Qwen3-14B	0.54	0.65	0.61	0.50	0.68
Qwen3-235B	0.41	0.56	0.49	0.38	0.65

E. RQ5: Ablation Study

Table XXVI gives the model-level ablation results for each BAIFT variant, allowing the effect of mutation, rendering, hybrid scoring, and prioritization to be inspected separately.

Table XXVII averages the ablation results across the three representative target LLMs, providing the summarized evidence used to quantify the contribution of each component in the main paper.

TABLE XXIV
RQ4 TOP-FIVE VULNERABLE GROUPS RANKED BY $VG(g)$.

Target LLMs	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5
GPT-5.4	elderly(age) / 0.63	disabled(ability) / 0.61	Muslim(religion) / 0.57	immigrant(culture) / 0.56	low-income(social) / 0.47
Gemini-3-Flash	elderly(age) / 0.65	disabled(ability) / 0.63	war victim(victim) / 0.61	low-income(social) / 0.58	Muslim(religion) / 0.53
Llama-3.1-8B	low-income(social) / 0.80	war victim(victim) / 0.75	disabled(ability) / 0.73	elderly(age) / 0.73	Muslim(religion) / 0.71
Llama-3.3-70B	disabled(ability) / 0.77	low-income(social) / 0.74	elderly(age) / 0.72	immigrant(culture) / 0.68	Muslim(religion) / 0.62
Mixtral-8×7B	low-income(social) / 0.76	war victim(victim) / 0.75	immigrant(culture) / 0.74	elderly(age) / 0.72	disabled(ability) / 0.65
DeepSeek-V3.2	elderly(age) / 0.71	low-income(social) / 0.68	disabled(ability) / 0.67	rural resident(social) / 0.61	immigrant(culture) / 0.61
Kimi-K2.5	low-income(social) / 0.73	elderly(age) / 0.69	disabled(ability) / 0.67	immigrant(culture) / 0.64	war victim(victim) / 0.64
GLM-5	rural resident(social) / 0.69	disabled(ability) / 0.65	elderly(age) / 0.64	low-income(social) / 0.61	immigrant(culture) / 0.57
Qwen3-14B	disabled(ability) / 0.84	low-income(social) / 0.78	war victim(victim) / 0.73	Muslim(religion) / 0.73	elderly(age) / 0.71
Qwen3-235B	elderly(age) / 0.72	disabled(ability) / 0.70	immigrant(culture) / 0.65	low-income(social) / 0.64	rural resident(social) / 0.59

TABLE XXV
RQ4 TOP-FIVE INTERSECTIONAL GROUP PAIRS RANKED BY $Amp(c)$.

Target LLMs	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5
GPT-5.4	<Black(race), Muslim(religion)> / 0.15	<female(gender), elderly(age)> / 0.11	<disabled(ability), overweight(body)> / 0.09	<immigrant(culture), crime victim(victim)> / 0.08	<low-income(social), caregiver(occupation)> / 0.07
Gemini-3-Flash	<disabled(ability), overweight(body)> / 0.12	<Arab(race), Muslim(religion)> / 0.11	<low-income(social), service worker(occupation)> / 0.10	<female(gender), elderly(age)> / 0.09	<immigrant(culture), abuse victim(victim)> / 0.08
Llama-3.1-8B	<immigrant(culture), war victim(victim)> / 0.18	<Black(race), Muslim(religion)> / 0.18	<female(gender), elderly(age)> / 0.17	<low-income(social), cleaner(occupation)> / 0.15	<disabled(ability), obese(body)> / 0.14
Llama-3.3-70B	<disabled(ability), overweight(body)> / 0.17	<low-income(social), delivery worker(occupation)> / 0.17	<Arab(race), Muslim(religion)> / 0.15	<immigrant(culture), crime victim(victim)> / 0.13	<female(gender), elderly(age)> / 0.12
Mixtral-8×7B	<disabled(ability), obese(body)> / 0.22	<immigrant(culture), disaster victim(victim)> / 0.17	<Black(race), Muslim(religion)> / 0.16	<low-income(social), caregiver(occupation)> / 0.15	<female(gender), elderly(age)> / 0.14
DeepSeek-V3.2	<female(gender), elderly(age)> / 0.13	<disabled(ability), overweight(body)> / 0.13	<low-income(social), cleaner(occupation)> / 0.13	<immigrant(culture), abuse victim(victim)> / 0.09	<Arab(race), Muslim(religion)> / 0.08
Kimi-K2.5	<immigrant(culture), war victim(victim)> / 0.20	<female(gender), elderly(age)> / 0.14	<Black(race), Muslim(religion)> / 0.14	<disabled(ability), obese(body)> / 0.12	<low-income(social), service worker(occupation)> / 0.11
GLM-5	<immigrant(culture), disaster victim(victim)> / 0.16	<disabled(ability), overweight(body)> / 0.14	<low-income(social), caregiver(occupation)> / 0.12	<Arab(race), Muslim(religion)> / 0.11	<female(gender), elderly(age)> / 0.09
Qwen3-14B	<disabled(ability), obese(body)> / 0.19	<female(gender), elderly(age)> / 0.18	<low-income(social), cleaner(occupation)> / 0.16	<Black(race), Muslim(religion)> / 0.13	<immigrant(culture), war victim(victim)> / 0.10
Qwen3-235B	<disabled(ability), overweight(body)> / 0.19	<female(gender), elderly(age)> / 0.18	<immigrant(culture), crime victim(victim)> / 0.17	<low-income(social), delivery worker(occupation)> / 0.14	<Arab(race), Muslim(religion)> / 0.12

TABLE XXVI
RQ5 RAW ABLATION RESULTS.

Target LLMs	Variant	#EDIs	#IDIs	IDI Ratio	$\overline{F_v(c)}$	$\overline{Amp(c)}$	Time/EDIs	Time/IDIs
GPT-5.4	BAIFT-full	548	807	0.60	0.55	0.13	103.51	70.29
GPT-5.4	w/o Mutation	474	648	0.58	0.51	0.11	111.84	81.81
GPT-5.4	w/o Rendering	464	539	0.54	0.48	0.08	128.62	110.72
GPT-5.4	w/o $s^{(r)}(\cdot)$	508	686	0.57	0.47	0.12	109.17	80.84
GPT-5.4	w/o $s^{(l)}(\cdot)$	497	678	0.58	0.45	0.12	110.43	80.95
GPT-5.4	w/o $s^{(j)}(\cdot)$	458	551	0.55	0.43	0.09	132.76	110.35
GPT-5.4	w/o Prioritization	524	762	0.59	0.56	0.13	158.34	108.88
DeepSeek-V3.2	BAIFT-full	1,238	1,381	0.53	0.61	0.12	89.47	80.21
DeepSeek-V3.2	w/o Mutation	976	909	0.48	0.56	0.10	96.83	103.97
DeepSeek-V3.2	w/o Rendering	1,014	945	0.48	0.50	0.07	112.94	121.19
DeepSeek-V3.2	w/o $s^{(r)}(\cdot)$	1,151	887	0.44	0.53	0.11	94.58	122.73
DeepSeek-V3.2	w/o $s^{(l)}(\cdot)$	934	860	0.48	0.52	0.11	95.21	103.40
DeepSeek-V3.2	w/o $s^{(j)}(\cdot)$	1,036	998	0.49	0.46	0.08	118.67	123.19
DeepSeek-V3.2	w/o Prioritization	1,154	1,193	0.51	0.60	0.12	139.72	135.15
Qwen3-14B	BAIFT-full	2,077	2,721	0.57	0.73	0.17	73.55	56.14
Qwen3-14B	w/o Mutation	1,812	2,214	0.55	0.67	0.15	79.46	65.03
Qwen3-14B	w/o Rendering	1,724	1,897	0.52	0.61	0.10	94.38	85.77
Qwen3-14B	w/o $s^{(r)}(\cdot)$	1,936	2,341	0.55	0.64	0.16	77.18	63.83
Qwen3-14B	w/o $s^{(l)}(\cdot)$	1,908	2,315	0.55	0.63	0.15	78.06	64.34
Qwen3-14B	w/o $s^{(j)}(\cdot)$	1,756	1,994	0.53	0.56	0.12	99.72	87.82
Qwen3-14B	w/o Prioritization	2,018	2,651	0.57	0.72	0.17	118.63	90.30

TABLE XXVII
RQ5 AVERAGED ABLATION RESULTS.

Variant	#EDIs	#IDIs	IDI Ratio	$\overline{F_v(c)}$	$\overline{Amp(c)}$	Time/EDIs	Time/IDIs
BAIFT-full	1,288	1,636	0.56	0.63	0.14	88.84	68.88
w/o Mutation	1,087	1,257	0.54	0.58	0.12	96.04	83.60
w/o Rendering	1,067	1,127	0.51	0.53	0.08	111.98	105.89
w/o $s^{(r)}(\cdot)$	1,198	1,305	0.52	0.55	0.13	93.64	89.13
w/o $s^{(l)}(\cdot)$	1,113	1,284	0.53	0.53	0.13	94.57	82.90
w/o $s^{(j)}(\cdot)$	1,083	1,181	0.52	0.48	0.10	117.05	107.12
w/o Prioritization	1,232	1,535	0.56	0.63	0.14	138.90	111.45